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The CDS-bond basis

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Abstract

We investigate the cross-sectional variation in the credit default swap (CDS)-bond bases and test explanations for the violation of the arbitrage relation between cash bond and CDS contract, which states that the basis should be zero in normal conditions. The evidence is consistent with "limits to arbitrage" theories in that deviations are larger for bonds with higher frictions as measured by trading liquidity, funding cost, counterparty risk, and collateral quality. Surprisingly, we find the basis to be more negative when bond lending fee is higher suggesting that arbitrageurs are unwilling to engage in a negative basis trade when short interest on the bond is high.

1 | INTRODUCTION

Financial markets experienced tremendous disruptions during the 2007–2009 financial crisis. Credit spreads across all asset classes and rating categories widened to unprecedented levels.[1] Perhaps, even more surprising, many relations that were considered to be textbook arbitrage prior to the crisis were severely violated. For example, in the currency markets, violations of covered interest rate parity occurred for currency pairs involving the US dollar (Coffey, Hrung, & Sarkar, 2009). In the interest rate markets, the swap spread that measures the difference between Treasury bond yields and Libor swap rates turned negative. In the interbank markets, basis swaps that exchange different tenor Libor rates (e.g., three months for six months) deviated from zero. In the inflation markets, break-even inflation rates turned negative implying an arbitrage with inflation swaps (Fleckenstein, Longstaff, & Lustig, 2014). In the credit markets, the credit default swap (CDS)-bond basis that measures the difference between CDS spreads and cash-bond implied credit spreads turned negative.

These anomalies suggest that such relations are not, in fact, arbitrage opportunities in the traditional textbook sense. Indeed, arbitrage profits may be difficult to realize in practice. Many of these relations involve a fully funded (e.g., cash) instrument and one or more unfunded derivative positions. Thus, counterparty risk of the derivative issuer may have rendered the arbitrage risky. Furthermore, funding cost differentials between cash instruments and derivative positions may have made the arbitrage costly to implement for an investor requiring funding. In the latter case, the arbitrage violations persist due to "limits to arbitrage" such as the inability of arbitrageurs to raise capital quickly and/or their unwillingness to take large positions in these arbitrage trades because of mark-to-market risk. These apparent arbitrage violations provide an interesting opportunity to test several of the limits to arbitrage theories (e.g., as surveyed by Gromb & Vayanos, 2010).

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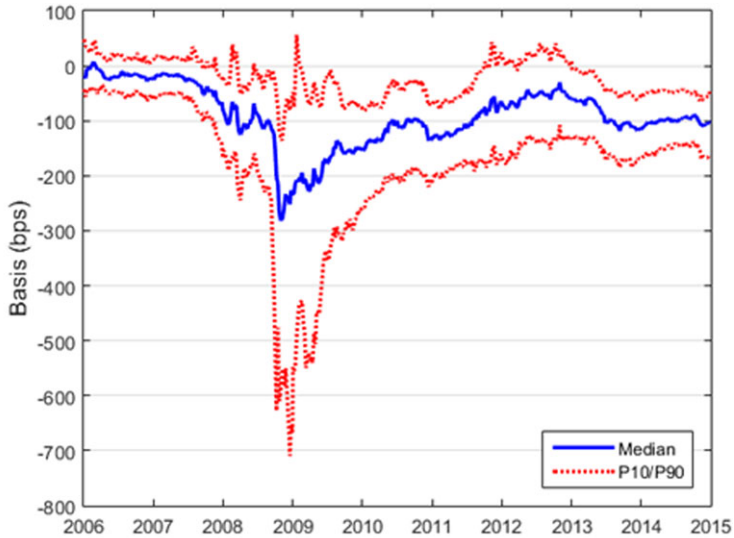
(a) Investment-grade Bonds



(b) High Yield Bonds

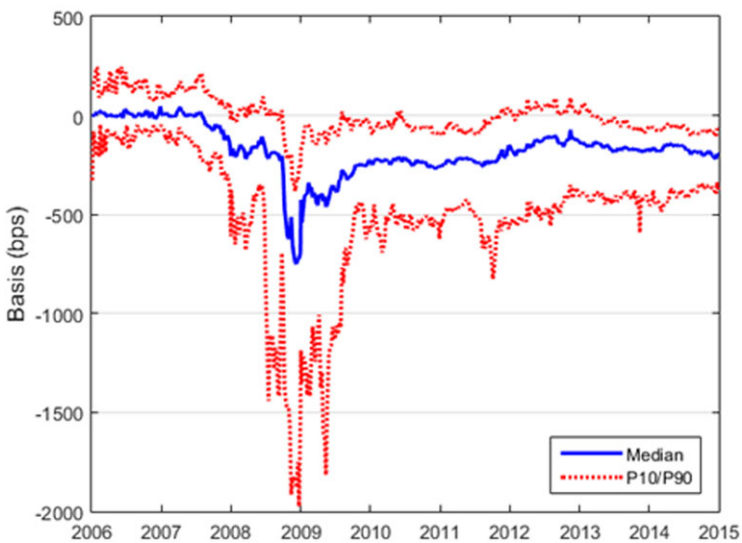


FIGURE 1 Dispersion of the CDS-bond basis by credit rating: (a) Investment-grade bonds and (b) high-yield bonds [Color figure can be viewed at wileyonlinelibrary.com]

In this paper, we focus on the CDS-bond basis that measures the difference between the CDS spread of a specific company and the credit spread paid on a bond of the same company. Figure 1 plots the time series of the CDS-bond basis for investment grade (IG) and high yield (HY) bonds. The plots indicate that the average basis for IG firms, which usually hovers around −17bps prior to the crisis, fell to −243 bps, and the average basis for HY firms dropped from 12 to −560 bps. In addition, the bases for both IG and HY firms remain negative even after the financial crisis. At first sight, a large negative basis smacks of arbitrage as it suggests that an investor can purchase the bond, fund it at Libor,

and insure the default risk on the bond by buying protection via the CDS contract. The resulting trade is virtually risk free yet, as the plots illustrate, it generates between 243 and 560 bps in arbitrage return per annum.

Studying the CDS-bond basis during the crisis is interesting for several reasons. First, early studies of this basis found that the arbitrage relation between CDS and cash-bond spreads holds fairly well during the precrisis period (Blanco, Brennan, & Marsh, 2005; Hull, Predescu, & White, 2004; Longstaff, Mithal, & Neis, 2005). In fact, if anything, these studies typically conclude that the basis should be slightly positive. Indeed, the arbitrage is, in general, not perfect (Duffie, 1999) and there are a few technical reasons (such as the difficulty in short selling bonds and the cheapest-to-deliver option) that tend to push the basis into the positive domain (Blanco et al., 2005). However, during the crisis, the bases were tremendously negative suggesting the need for alternative explanations.

Second, there is a large cross-sectional variation in the observed bases across individual firms. This cross-sectional variation makes the basis arbitrage an interesting laboratory to test various limits-to-arbitrage theories. It is the focus of our paper.

There are several reasons why one might expect the basis to become negative during the financial crisis of 2007–2009. Anecdotes for the negative basis claim that several major financial institutions, pressed to free up their balance sheet and to improve their cash balance, reduced their leverage by selling off bonds. The argument is that deleveraging exerted downward pressure on bond prices and thus upward pressure on credit spreads relative to CDS spreads that represent the "fair" value of the default risk insurance. However, this cannot be the whole story. In a perfect frictionless market, investors would simply borrow cash to buy the bonds, buy protection, and finance the position until maturity or default. For deleveraging to have a persistent impact on the basis, there must be some limits to arbitrage (Shleifer & Vishny, 1997). In particular, if risk capital is limited, then mark-to-market basis trade becomes risky and investors will tend to buy the bonds-basis packages that are (ex-ante) most attractive from a risk-return trade-off.

In this paper, we analyze the risk-return trade-off in a basis trade for an investor with limited capital. We find that the investor is exposed to: (a) increased collateral values of the underlying bonds, (b) increased illiquidity of bond trading, (c) counterparty risk in buying CDS contracts, and (d) increased funding constraints faced by investors who want to explore the arbitrage. For a given level of the basis, we expect investors with limited arbitrage capital to prefer those trades with better collateral, higher trading liquidity, less counterparty risk, and lower exposure to funding costs. Put differently, if arbitrageurs have limited capital, then, in equilibrium, we expect to see a larger basis (which can be thought of as an expected return) for firms with poorer collateral quality, lower trading liquidity, more counterparty risk, and higher funding liquidity risk. To disentangle which explanation is most relevant in explaining the deviations in the CDS-bond basis, we construct measures of collateral quality, bond liquidity risk, counterparty risk, and funding liquidity risk for each firm and explore whether they can explain the cross section of the bases for individual bonds and for a set of 24 bond portfolios sorted on rating and size.

We find that all of our limits to arbitrage measures matter and together they explain about 80% of the cross-sectional variation in the bases of 24 size and rating sorted portfolios and about 40% of the cross-sectional variation of individual firm bases. During the crisis when the basis became the most negative, all limit-to-arbitrage measures, collateral quality, bond illiquidity, counterparty risk, and funding liquidity risk, are statistically significantly correlated with the basis and have the expected sign, suggesting that the basis was more negative for a bond more costly engaged in the arbitrage trade. We find one exception. The bond lending fee is negatively associated with the negative bond basis. This is surprising since a high lending fee is typically associated with higher collateral value and should make the arbitrage trade easier to implement (i.e., since the negative basis arbitrage involves buying the bond, the lending fee could be earned by the arbitrageur). In contrast, and as expected, bond lending fees are associated positively with the basis when the bond basis is positive (i.e., a higher lending fee indicates that the bond is costly to short rendering the positive basis arbitrage trade more costly). We speculate that arbitrageurs refrain from engaging in a negative basis trade with a high fee bond as the high lending fee proxies the high short interest on the bond (Bai, 2018; Reed, 2013) and the greater demand in the short selling arbitrage signals that bond prices may drop further and, as such, the basis could further widen. Thus, arbitrageurs could be avoiding a "catching-a-falling-knife" trade. In fact, we find that for firms with a negative basis, the basis tends to be more negative if the underlying bond has poorer ratings, higher CDS, and a smaller size. All of these characteristics typically correlate negatively with collateral quality. For firms with a negative

basis, the lending fee seems to capture different information than collateral quality. In sum, our cross-sectional results strongly support the hypothesis that limits to arbitrage prevent arbitrageurs from closing the basis gap.

The negative basis has attracted considerable attention in the practitioner literature (i.e., D.E. Shaw Group, JP Morgan "The bond-CDS basis handbook," 2009, Mitchell & Pulvino, 2012). These papers emphasize the role of financing risk in generating the negative basis, as well as deleveraging in generating downward price pressure on cash bonds. In the academic literature, Garleanu and Pedersen (2011) provide a theoretical model, where leverage constraints can generate a pricing difference between two otherwise identical financial securities that differ in terms of their margin requirements or haircuts. Specifically, their theory predicts that the difference between two bases should be related to the difference in margin requirements (i.e., haircuts) times the difference between the collateralized and uncollateralized borrowing rate. Our study differs from previous papers in that we focus on the cross-sectional variation in individual firm bases (rather than on the average basis level) and try to relate it to firm, bond, and CDS characteristics. Two contemporaneous papers have also investigated the CDS-bond basis. Fontana (2011) studies the time-series variation in the average basis using cointegration techniques. Kim, Li, and Zhang (2016, 2017) examine a long–short basis risk factor and study its implications on corporate bond returns.

The paper proceeds as follows. The next section discusses practical issues regarding an actual basis trade and isolates the various sources of risk in such a trade. Section 3 introduces the data. In Sections 4 and 5, we construct limits-to-arbitrage measures and examine their impact in explaining the cross-sectional variation of the bases, while Section 6 provides our conclusions.

2 | THE CDS-BOND BASIS

A CDS is essentially an insurance contract against a credit event of a specific referenced entity. It is an over-the-counter transaction between two parties in which the protection buyer makes periodic coupon payments to the protection seller until maturity or until some credit event occurs. When a credit event occurs, the protection buyer delivers a bond from an eligible pool to the protection seller in exchange for its par value.[2]

The contract is designed so that the owner of a particular bond can hedge their credit risk exposure to the issuer of that bond by buying CDS protection on that counterparty. As a result, we would expect CDS spreads to be similar to the credit spreads observed on corporate bonds that are deliverables of the CDS contract. In fact, under some conditions, an exact arbitrage relation exists, implying that the CDS spread should equal the credit spread on the deliverable corporate bond.[3] This leads to the theoretical definition of the CDS-bond basis as the CDS spread minus the corporate bond credit spread.

While the CDS spread is observable in the market, it is not obvious how to compute the appropriate corporate bond spread. As discussed by Duffie (1999), the ideal corporate bond spread would be the spread over Libor of a floating rate note with the same maturity as the CDS referenced on the same firm. In practice, this spread is often not observable as firms rarely issue floating rate notes. Instead, we must rely on other available fixed rate corporate bond prices. Several methodologies have been proposed in the literature. Following Elizalde, Doctor, and Saltuk (2009), we adopt the par equivalent CDS (PECDS) methodology. This method, which we present for completeness in Appendix A, essentially amounts to extracting the default intensity consistent with the prices of the corporate bonds observed in the market and using the Libor swap curve as the risk-free benchmark curve. Then, one can calculate the fair CDS spread consistent with the bond implied default intensity and the risk-free benchmark curve given a standard recovery assumption. It is this theoretical bond-implied CDS spread, called the PECDS spread, that we compare to the quoted CDS spread on the same reference entity to define the CDS-bond basis:

$$Basis_i(\tau) = CDS_i(\tau) - PECDS_i(\tau), \tag{1}$$

where τ is the maturity and i indicates the reference entity. This methodology has several advantages as reviewed in Elizalde et al. (2009). It has also been used by previous academic studies, such as Nashikkar, Subrahmanyam, and

Mahanti (2011). Another important issue in the measurement of the basis is the funding or risk-free rate benchmark (Hull et al., 2004).

Several authors have argued that the Treasury curve is not the appropriate risk-free benchmark and that it is lower than the typical funding cost an investor can achieve via collateralized borrowing (see Longstaff, 2004 and Feldhütter & Lando, 2008 among others). In fact, Hull et al. (2004) use the basis package (a portfolio long several corporate bonds and long CDS protection) to define a risk-free asset available to any investor. They argue that since the average CDS-bond basis is zero when measuring funding costs using the swap rate minus 10 bps and the CDS-bond basis exhibits little cross-sectional variation, this is evidence that the "true" shadow risk-free rate for a typical investor is around the swap minus 10 bps (or approximately the Treasury plus 50 bps).

We emphasize that the very large cross-sectional variation in the basis (across rating categories) documented in Figure 1 allows us to immediately dismiss the fact that mismeasurement of the risk-free rate benchmark is the explanation for the puzzling behavior of the CDS-bond basis during the crisis. If we were simply mismeasuring the risk-free benchmark, we would find an approximately constant CDS-bond basis across firms reflecting the spread between our benchmark risk-free curve and the true (unobserved) risk-free curve. However, since we do not observe the true risk-free benchmark curve, it is important to focus on the cross-sectional variation in the basis, rather than focusing on the average level, which could be affected by the "flight-to-quality" effect documented in the Treasury and swap literature.

When the basis is positive, the CDS spread is larger than the bond spread. An investor could then short the bond and sell the CDS protection to capture the basis. When the basis is negative, the CDS spread is lower than the bond spread. By buying the bond and buying CDS protection, investors could lock in a risk-free annuity equal to the absolute value of the basis.

As discussed in the introduction, during normal times, the CDS-bond basis tends to be very small and, if anything, slightly positive. This has been studied extensively by Blanco et al. (2005), Longstaff et al. (2005), and recently by Nashikkar et al. (2011). However, Figure 1 reveals that the CDS-bond basis was significantly and persistently negative during the financial crisis and after the crisis. Furthermore, there was substantial cross-sectional variation in the negative basis as noted by the conspicuous difference in the bases between IG and HY bonds.

While a positive basis can often be traced back to some inability to implement the arbitrage trade because either the bonds are difficult to short (see liquidity concerns in Nashikkar et al., 2011) or there exists a cheapest-to-deliver option (see Blanco et al., 2005), a negative basis is harder to explain. In the negative basis case, the arbitrage trade requires buying the bond, financing its purchase, and buying protection to hedge against the default event. Figure 2 suggests that the return to the "negative basis" trade would have been between 243 and 560 bps for IG and HY bonds, respectively. These seem like strikingly arbitrage profits. Thus, it is important to review the details of such a basis trade implementation to better understand where the limits to arbitrage may arise.

2.1 | Negative basis trade

In practice, there are several reasons why a negative basis trade is not a pure arbitrage. These risks are discussed in detail in Elizalde et al. (2009) (see, in particular, their table 2 on page 23). The main issues when implementing a negative basis trade have to do with funding risk, sizing the long CDS position, liquidity risk, and counterparty risk.

Suppose we find a bond with a negative basis that trades at a price B below its notional of N. A negative basis trade requires buying the bond. The purchase is funded via the repo market where investors face a haircut h. This effectively implies that arbitrageurs will have to provide hB dollars of "risk capital" funded at $Libor + f$ where f is the funding spread over Libor faced by the arbitrageur. The repo contract is typically overnight (up to a few months at most) with an agreed upon repo rate, and needs to be rolled over repeatedly until the maturity of the basis trade, which is the lesser of the default and the maturity (e.g., five years).

At the same time, the investor buys the protection in the CDS market to offset the default risk. A question arises as to how to size the CDS position. A conservative approach from the viewpoint of minimizing exposure to a jump to default is to buy protection on the full notional N of the bond.

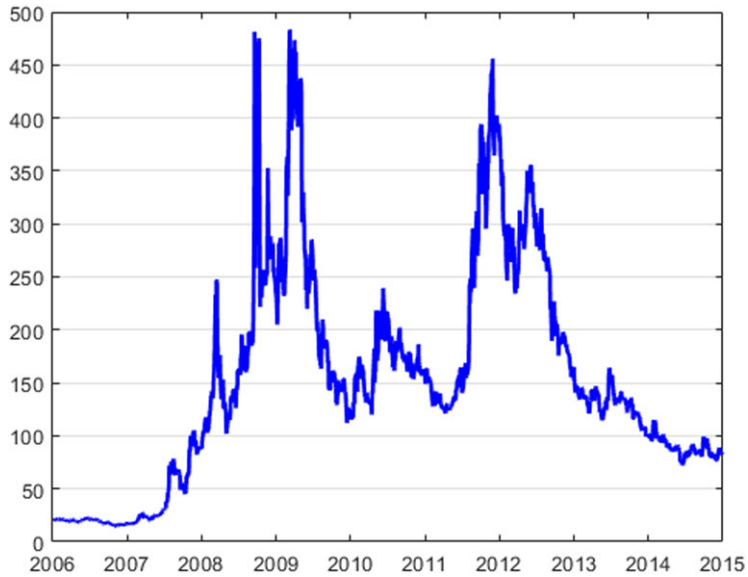


FIGURE 2 The Average Five-Year CDS Spread for the Primary Dealers (bps) [Color figure can be viewed at wileyonlinelibrary.com]
Source: Markit Inc. and Federal Reserve Bank of New York.

Market participants typically prefer to buy less protection to improve the carry profile of the trade (pay less in insurance premiums). The justification is that the maximum capital at risk in the transaction is the initial purchase price of B (for bonds that trade at a premium, one may buy more protection than the nominal). In fact, a customary approach is to make an assumption about recovery (i.e., assume that in the case of bankruptcy, a fraction R of the notional of the bond is recovered) and buy protection on a CDS notional of N_{CDS} so as to cover the loss in capital (i.e., such that $B - N\,R\ = N_{CDS}\ (1 - R)$). This will increase the carry of the trade as the CDS premia are now reduced, but expose the investor to a jump to default in case the recovery is smaller than expected. An alternative approach is to choose the notional of the CDS position to match the spread duration on the risky bond. This approach tries to minimize the mark-to-market differences between the bond and the CDS position over the life of the bond as opposed to thinking about the jump-to-default risk. As explained in Duffie (1999), there is no perfect arbitrage when the underlying bond is not a floating rate note with the same maturity as the CDS contract. Even in that case, the arbitrage is not perfect (Lando, 2004).

For illustration, suppose the investor buys protection on a notional N_{CDS}. This requires a margin payment of M and periodic mark-to-market margin calls. The margin must be funded at $Libor + f$.[4] After one day, the profit or loss (P&L) on the trade can be written as:

$$\begin{aligned} P\&L\,(t+1) \ &= B_{t+1}^{bid} - B_t^{ask} + N_{CDS} D_{CDS} * \left(CDS_{t+1}^{bid} - CDS_t^{ask} \right) \\ &\quad - B_t^{ask} * \left[h\,(Libor + f) + (1 - h) * (Repo) \right] - M_t\,(Libor + f)\,, \end{aligned} \tag{2}$$

where D_{CDS} is the duration of the CDS such that the P&L on the CDS is the product of the duration with the change in the CDS rate. Note that if CDS increases, the short credit/long protection position makes money. In the P&L analysis, we explicitly consider the trading costs in the form of bid–ask spreads on the bond and the CDS contract. For example, suppose we size our position in the CDS contract to match the Libor spread duration on the corporate bond. Then, we can rewrite the P&L as[5]

$$\begin{aligned} P\&L\,(t+1) &\approx D_B * \Delta Basis_t - D_B \Delta BAS_t \\ -B_t^{ask} * &\left[h\,(Libor + f) + (1 - h) * (Repo) \right] - M_t\,(Libor + f)\,. \end{aligned} \tag{3}$$

BAS is the average of the bond yield bid–ask spread and the CDS bid–ask spread. Specifically, this relation shows that a typical basis trade, when rolled over repeatedly, is exposed to

- an increase in funding costs as measured by the benchmark Libor rate.
- an increase in the arbitrageur's own credit risk, which could lead to a larger markup (f). We note that if the arbitrageur has a large position in basis trades, then this could be tied to the basis becoming more negative (i.e., the trade running away from him).
- a worsening of the collateral quality of the bond that could lead to an increase in the haircut (h) and the repo rate (*Repo*).
- an increase in the margin requirements on the CDS position (M_t).
- an increase in trading costs as measured by the average bid-ask spread on the bond and CDS.

This component is important for two reasons. First, it affects the daily mark-to-market positions. In addition, if the position does need to be unwound prior to the maturity of the contract (or default), trading liquidity matters.

Finally, the trade is also affected by counterparty risk in the sense that if a default on a bond occurs at time $(\tau)_B$, then the P&L will be:

$$P\&L\left(\tau_B\right) \;=\; R\,N + N_{CDS}\,(1-R)\,1_{\tau_C > \tau_B}, \tag{4}$$

where τ_C denotes the default time of the counterparty selling protection and R denotes the realized recovery on the bond. Specifically, if the counterparty defaults (or has defaulted) when the underlying firm defaults, then the CDS protection expires. This highlights the fact that from an ex ante perspective, counterparty risk depends upon the correlation between the default risk of the underlying name and the counterparty selling the protection. This is typically a large bank, such as J.P. Morgan, Lehman Brothers, or Goldman Sachs.

It is important to stress that counterparty risk is typically viewed as likely to be small if the counterparty defaults prior to the default event (i.e., $\tau_C < \tau_B$) and, if the marking to market were perfect, then the investor could reopen a new position at no cost with another counterparty the same day as the underlying bond ($\tau_C = \tau_B$). In practice, however, it is likely that the failure of the counterparty, especially during an extraordinary period like a financial crisis, would be associated with more substantial costs and risks for the investor. These losses would typically be related to the likely mark-to-market loss in the position on the day of the counterparty default. They may also be related to technical considerations that have to do with the specific bankruptcy provisions in the International Swap and Derivative Association (ISDA) covering the CDS trade (e.g., if the mark-to-market limits were insufficient, if the collateral posted with the counterparty was rehypothecated, or if the cash settlement upon bankruptcy of the counterparty is based on mid-market quotes).

Below, we attempt to use the cross-sectional variation in individual bond bases to disentangle the effects of the various risks outlined above that affect the risk-return trade-off of a basis trade. Our working hypothesis is that an arbitrageur with limited access to capital will try to exploit the basis trade opportunities that offer the best expected return per unit of risk capital. Thus, they will choose basis trades that have the most negative basis (highest expected return) controlling for ex ante measures of exposure to market and funding liquidity. All else being equal, they will prefer basis trades on bonds with lower haircuts, lower exposure to funding costs (i.e., in the sense that for two bonds with equally negative basis, the one that correlates more with funding costs is more attractive as the basis trade converges when funding costs rise), lower counterparty risk (i.e., in the sense that the probability of the underlying firm defaulting at the same time as the counterparty in the CDS is lower), and lower trading liquidity risk (i.e., lower current transaction costs and lower future risk-adjusted transaction costs). If this hypothesis is correct, then we expect that the risk characteristics of the basis trade (e.g., counterparty risk, funding risk, liquidity risk, and collateral quality) should be related to the cross-sectional variation of the bases.

3 | DATA

The data used to study the CDS-bond basis come from several sources. We begin with the universe of firms whose single-name CDS is traded in the derivatives market and their transactions are recorded in the Markit database. Then, we identify corporate bonds issued by these firms from the Mergent Fixed Income Database and collect bond characteristics. We further download corporate bond transaction records from the enhanced version of the Trade Reporting and Compliance Engine (TRACE). Finally, we match each firm's CDS and bond spread to the corresponding equity returns in the Center for Research in Security Prices (CRSP). All data are in daily frequency from July 1, 2006 to December 30, 2014. The entire sample is further partitioned into four phases. Phase 1 is the period prior to the subprime credit crisis named "Before Crisis" (7/1/2006-6/30/2007); Phase 2 is the period between the subprime credit crisis and the bankruptcy of Lehman Brothers called "Crisis I" (7/1/2007-8/31/2008); Phase 3 is the period after the Lehman Brothers' failure, "Crisis II" (9/1/2008-9/30/2009), while and Phase 4 is the period after the financial crisis, "Postcrisis" (10/1/2009-12/30/2014).[6]

Our goal is to examine the arbitrage relationship between corporate bond cash and the derivatives market. The cornerstone of our analysis is an accurate measure of the CDS-bond basis. As demonstrated in Section 2 and Appendix A, the basis is constructed from the CDS spread, the corporate bond transaction price, and the reference interest rate.

3.1 | Corporate bonds

Corporate bond transaction data are downloaded from TRACE. Bond characteristics are collected from Mergent Fixed Income Databases including coupons, ratings, interest rate frequency, option features, and so on. For bond-level rating information, if a bond is rated only by Moody's or by Standard and Poor's, we use that rating. If a bond is rated by both rating agencies, we take the average rating as the final one.

We highlight the following filtering criteria in order to choose qualified bonds. First, we remove bonds that are not listed or traded in the US public markets including bonds issued through private placement, bonds issued under the 144A rule, bonds that do not trade in US dollars, and bond issuers not in the jurisdiction of the United States. In addition, we focus on corporate bonds that are not structured notes, not mortgage backed, or asset backed. We also remove bonds that are agency-backed or equity-linked. Moreover, we exclude convertible bonds as this option feature distorts the basis calculation and makes it impossible to compare the basis of convertible and nonconvertible bonds.[7] Finally, we remove bonds with floating rates. As such, the sample is composed of only bonds with fixed or zero coupons. This rule is applied based on the consideration of the accuracy in the basis calculation given the challenge in tracking floating coupon bonds's cash flows.

The enhanced TRACE provides information regarding bond transactions at the intraday frequency. Beyond the above filtering criteria, we further clean up the TRACE transaction records by eliminating when-issued bonds, locked-in bonds, and bonds with commission trading, special prices, or special sales conditions. We remove transaction records that are cancelled, and adjust records that are subsequently corrected or reversed. Bond trades with more than two-day settlements are also removed from our sample.

Finally, we focus on the bonds that have three to seven and one-half years remaining to maturity (time-to-maturity is measured each day during the sample period). This criterion is due to the concern that in the CDS market, which we introduce in the next subsection, five-year CDS contracts often have the best liquidity. To match the five-year term, we limit the calculation of the CDS-bond basis only for those bonds with time-to-maturity ranging from three to seven and one-half years.

3.2 | Credit default swap

We download single-name CDS data from Markit Inc. for US firms. The prices are quoted in basis points per annum for a notional value of $10 million and are based on the standard ISDA contract for physical settlement. The original data set provides daily market CDS prices in various currencies and different types of restructuring documentation clauses.

Following a conventional rule, we choose the CDS price in US dollars and the documentation clause type as "modified restructuring" (MR).[8]

There are several caveats in matching CDS with underlying reference entities. First, the MR rule restricts deliverable bonds to be within 30 months of contract maturity. We check the bond maturity to follow the MR rule. In addition, the underlying bond should be deliverable into the CDS contract. However, it is very difficult to identify whether bonds are deliverable into CDS contracts for a large sample of firms over a long time period "since CDS conventions are often bilaterally defined in the over-the-counter market," as pointed out by Nashikkar et al. (2011).

The original data set provides a term structure of CDS spreads incorporating maturities of 1, 2, 3, 4, 5, 7, and 10 years. We use all maturities in conjunction with matching interest rate swaps to calculate a term structure of default probability. This is an integral component in deriving the bond-implied CDS spread (PECDS) and the CDS-bond basis (see Appendix A). In the end, we focus on the CDS-bond basis with a maturity of five years as the five-year CDS is by far the most liquid in the credit derivatives market, and this is also the one primarily used in the literature.

The CDS data and corporate bond data are manually matched via company names. There are originally 1,842 unique CDS underlying entities from 2006 to 2014. After matching to the corporate bond data and applying the aforementioned filtering criteria, we are left with a total of 1.1 million daily observations from July 1, 2006 to December 31, 2014 representing 4,415 pairs of CDS-bond basis for 679 unique CDS names. To reduce the influence of outliers in the cross-sectional regressions, we further winsorize the bases at the 0.5% and 99.5% level. Our empirical results in Section 5 are similar if winsorizing at the 1% and 99% level or at the 0.25% and 99.75% level.

3.3 | Reference rate

We use the US dollar interest rates swaps as reference rates for the risk-free funding curve for computing credit spreads. An alternative choice might be to use government bond yields. However, as Blanco et al. (2005) point out, "government bonds are no longer an ideal proxy for the unobservable risk-free rate" due to tax treatment, repo specialness, legal constraints, and other factors. More importantly, the Libor swap rate represents a better indicator of the funding costs for financial intermediaries and typical basis swap traders than the Treasury curve. Therefore, we use it as our benchmark funding curve for the basis calculations (see also the swap-Treasury spread discussions in Collin-Dufresne & Solnik, 2001; Hull et al., 2004; Longstaff et al., 2005).

As discussed in Section 2, we focus on the cross-sectional variation in the CDS-bond basis rather than its absolute level since we do not observe the true risk-free reference rate.

3.4 | Summary statistics

Table 1 presents the summary statistics of the CDS-bond basis. The basis across all firms was slightly negative prior to the crisis, −10 bps, on average, from 7/1/2006 to 6/30/2007 (consistent with the evidence in Hull et al., 2004), but fell to −118 bps in the first phase of the financial crisis and to −324 bps after the bankruptcy of Lehman Brothers. The average basis remained negative around −137 bps after the financial crisis. Meanwhile, the volatility of the basis kept increasing for all types of firms from an average of 59 bps before the crisis to 192 bps and further to 369 bps during the turmoil of the financial crisis. The volatility fell back to 152 bps after the crisis, still far from the precrisis level. Firms with both IG and HY ratings share the same pattern as firms overall whose bases became more negative and volatile as the financial crisis progressed. Moreover, the basis of HY firms is always more volatile than that of IG firms. Financial firms and nonfinancial firms share similar time-series patterns in their bases except that financial firms have slightly higher volatility of basis during the peak of a financial crisis.

Table 1 also provides additional basis results across ratings from AAA/AA to CCC. Firms with lower credit ratings tend to have more negative and volatile bases during the crisis, while the bases display a right-skewed "smile" from AAA/AA to CCC before and after the crisis.

Figure 1 provides an illustration of the basis dynamics for IG and HY bonds. The solid blue line is the median value of the aggregated CDS-bond bases for bonds in each rating category, weighted by bond outstanding amounts. The

TABLE 1 Summary statistics of discrepancies in CDS and cash bond spreads

	Before Crisis				Crisis I				Crisis II				Post-crisis			
	July 2006–June 2007				July 2007–Aug. 2008				Sept. 2008–Sept. 2009				Oct. 2009–Dec. 2014			
	Mean	*SD*	P10	P90	Mean	*SD*	P10	P90	Mean	*SD*	P10	P90	Mean	*SD*	P10	P90
ALL	−10	59	−57	45	−118	192	−273	14	−324	369	−667	−55	−137	152	−268	−32
IG	−17	30	−51	17	−83	108	−150	−10	−243	256	−451	−48	−101	71	−173	−32
HY	12	104	−107	142	−180	265	−486	57	−560	504	−1,248	−114	−237	242	−477	−35
F	−24	26	−55	3	−117	161	−343	12	−351	450	−913	−4	−116	102	−205	−27
NF	−6	65	−56	59	−119	196	−254	15	−313	321	−625	−71	−145	165	−294	−34
AAA/AA	0	34	−31	51	−35	61	−90	32	−111	84	−218	−12	−68	40	−121	−23
A	−15	29	−49	17	−88	107	−154	−20	−187	180	−359	−32	−86	58	−145	−27
BBB	−23	29	−56	11	−103	68	−173	−37	−357	302	−607	−135	−122	77	−202	−49
BB	−7	40	−52	43	−107	126	−231	18	−493	313	−852	−182	−195	127	−363	−54
B	13	90	−104	125	−190	249	−499	62	−501	486	−1,178	−72	−221	205	−445	−29
CCC	27	151	−128	167	−296	398	−807	120	−1,113	568	−1,747	−534	−413	418	−951	−8

Notes: This table provides the descriptive statistics for the average CDS-bond basis in four phases. Phase 1 is the period prior to the subprime credit crisis, "Before Crisis" (July 2006–June 2007), Phase 2 is the period between the subprime credit crisis and the bankruptcy of Lehman Brothers, "Crisis I" (July 2007–August 2008), Phase 3 is the period after Lehman Brothers' failure, "Crisis II" (September 2008–September 2009), and Phase 4 is the period after the financial crisis, "Post-crisis" (October 2009–December 2014). The basis is calculated as the difference between the CDS spread and the par equivalent corporate bond spread using the methodology in the Appendix. The summary statistics are reported for all bonds (ALL), investmeng-grade bonds (IG), high-yield bonds (HY), bonds issued separately by financial firms (F) and nonfinancial firms (NF), as well as across rating categories: AAA/AA, A, BBB, BB, B, and CCC. We calculate the cross-sectional mean, standard deviation, the 10th and the 90th percentile value of the bases across all bonds each day, and report the time-series average of these statistics. All entries are in basis points.

dotted red lines are the 10th and 90th percentile of the aggregated bases. It is worth noting that the average CDS-bond basis for both the IG and HY bonds after the financial crisis, though improved, is still far below their precrisis levels. Moreover, there exists a large dispersion of the basis values throughout the entire sample period. We focus the analysis on the cross-sectional variation in the negative basis as the basis sample became predominantly negative. In the entire sample, 92.7% of the observations have negative bases; the proportion is even larger during the crisis period and the postcrisis period for about 94.4% and 93.9%, respectively.

3.5 | Preliminary evidence on the cross-sectional variation in the basis

Garleanu and Pedersen (2011) find that haircuts are typically around 25% for IG firms (and very similar across firms rated from AAA to BBB) and 55% for HY bonds (rated BB or lower). In their model, the basis differential between IG and HY bonds should be equal to the difference between the haircut margins multiplied by the collateral funding spread (i.e., the difference between the collateralized and the uncollateralized funding rates). While haircut is a plausible determinant of the basis, our data suggest that there exist additional important factors. As clearly illustrated in Table 1, there is tremendous amount of variation in the basis within a credit rating category and many differences in the basis within the IG and the HY categories.[9]

To illustrate this point even more dramatically, we present in Table 2 the examples for 12 firms in our sample that have a positive basis for more than 100 days during the second phase of the crisis (September 1, 2008–September 30, 2009 with 271 days). These firms have diverse credit ratings ranging from B (Las Vegas Sands Corp and Penn Natl Gaming Inc.) to AAA (Berkshire Hathaway and GE), and belong to six separate industries. This is clearly at variance with a model that would have a single factor, such as haircuts or margins, to explain the basis.[10] The haircut and margin requirements on Las Vegas Sands were much larger than for the Berkshire bonds, yet both display a positive

TABLE 2 Examples of firms with large positive basis during the crisis

Firm	Number of days with positive basis		Rating	Industry
	Crisis I ($T = 295$)	Crisis II ($T = 271$)		
Newmont Mng Corp	286	250	BBB	Basic Materials
Berkshire Hathaway	127	244	AAA	Financials
Amern Tower Corp	237	226	BB	Technology
Emc Corp	259	188	BBB	Technology
MetLife Insurance Co	12	178	A	Financials
Boyd Gaming Corp	253	163	BB	Consumer Services
General Electric Co	89	154	AAA	Industrials
Windstream Corp	54	131	BB	Telecommunications
Penn Natl Gaming Inc	134	130	B	Consumer Services
Mylan Inc	204	122	BB	Health Care
AutoNation Inc	1	117	BB	Consumer Services
Las Vegas Sands Corp	108	106	B	Consumer Services

Notes: This table shows examples of 12 firms with large number of positive days during the risis. We report the number of days with positive basis in Crisis I and Crisis II, as well as firm ratings and firm industries. Ratings are based on the values at the end of September 2008.

basis (when most IG and HY bonds displayed strongly negative bases at the time). Some factors driving the individual basis are likely to be highly idiosyncratic. For example, it was suggested to us that the very positive basis on Berkshire was due to the large demand from CDS protection buyers by dealers who had non-mark-to-market in-the-money long-term volatility exposures to Berkshire. This would have driven the CDS on Berkshire up relative to the bond yield generating the positive basis. We focus more systematically on the cross-sectional variation in the CDS-bond basis below.

4 | LIMIT-TO-ARBITRAGE FACTORS

The deviation between the same underlying bond's transaction price and its CDS spread, according to Section 2, could be driven by four types of factors: (a) the collateral value of the underlying corporate bond, (b) the liquidity of bond trading, (c) the counterparty risk in buying a CDS contract, and (d) the funding constraints faced by investors who want to explore the arbitrage. In this section, we construct various proxies for these factors and discuss their likely relationship to the cross section of the CDS-bond bases.

4.1 | Collateral quality proxy

In the basis trade, the purchase of a bond is funded via the repo market and two key variables determine the cost of the purchase: haircut and repo rate. We label these as the proxies of collateral quality. Haircuts of corporate bonds in the repo market often have a narrow range in the IG or non-IG categories. That is, bonds have similar haircuts across ratings from AAA to BBB or across ratings from BB and CCC. Garleanu and Pedersen (2011) find that haircuts are typically around 25% for most IG firms and around 55% for most non-IG firms. This rough haircut convention cannot help distinguish the wide range of the bases, as shown in Figure 1. Given that haircut is primarily driven by a bond's credit risk, we use credit ratings as one proxy for collateral quality, noted as *Rat*. In detail, we assign numeric values for bond ratings ranging from one for CCC and two for CC—all the way to 21 for AAA-rated bonds. As such, we have a

granularity in capturing the collateral quality of bonds. The higher the rating indicator, the higher the expected collateral quality.

An alternative proxy for the collateral value is the repo rate, but the bond-level data on the repo rates are not publicly available. Thus, we rely on the bond lending fee as a second proxy of collateral value based on unique corporate bond loan data. The data, provided by Markit, record daily corporate bond loan transactions where beneficial owners, such as insurance companies, pension funds, and other institutional investors, lend out bonds to end users, such as hedge funds, and collect lending fees. We collect the real borrowing cost for each bond in our sample, *Fee*, as the transaction value-weighted average lending fees over all open transactions. The lending fee is typically negatively correlated with the repo rate of a bond and positively related to the bond's collateral quality. It is a source of additional return for bondholders. Alternatively, a high lending fee signals a higher shorting demand for a particular bond and it can also signal a negative view on the bond by market participants (Bai, 2018).

To complete a negative basis trade, an arbitrageur must buy bonds that are funded via the repo market using the same bonds as collateral. The haircut and the repo rates imposed on the repo transaction reduce the amount of leverage available to the arbitrageur. All else being equal, we expect bonds with lower collateral quality (i.e., a higher haircut or a lower lending fee) to have a less profitable basis trade per unit use of expected risk capital. Thus, the lower the collateral quality, the more negative the basis to equalize the expected returns per unit of risk capital. As such, we expect a positive coefficient in the cross-sectional regressions of the bases on collateral quality.

4.2 | Bond trading liquidity

As the basis trade P&L analysis in Section 2.1 reveals, individual bond (and CDS) trading costs affect the profitability of the CDS-bond basis trade. Therefore, all else being equal, arbitrageurs will seek basis trades with bonds that are more liquid and that have less (trading) liquidity risk in the sense that they are less likely to become illiquid when the basis trade further diverges. Recent studies, such as Bao, Pan, and Wang (2011) and Dick-Nielsen, Feldhütter, and Lando (2012), find severe deterioration of liquidity in the corporate bond market during the financial crisis. In this section, we construct three different measures of bond liquidity and liquidity risks following the decomposition method proposed in Acharya and Pedersen (2005).

The bond illiquidity level (noted as *Liq*) is measured by the gamma (L^{BPW}) in Bao et al. (2011) that seeks to extract the transitory component in the bond price.[11] Specifically, let $\Delta p_t = p_t - p_{t-1}$ be the daily log price change from $t-1$ to t. The L^{BPW} is defined as:

$$L^{BPW} = -Cov_t\left(\Delta p_t, \Delta p_{t-1}\right). \tag{5}$$

We compute L^{BPW} for each bond per month and assign the monthly value for weeks within that particular month. We expect bonds that are more illiquid to have a more negative basis. That is a negative coefficient of the bases on the *Liq* measure in the cross-sectional regressions.

The bond iliquidity beta (denoted as β_{Liq}) is the comovement between bond i's illiquidity and the market illiquidity, where the market illiquidity is calculated as the average of individual bond illiquidity in the sample. As explained in Acharya and Pedersen (2005), investors want to be compensated for holding a security that becomes illiquid when the market, in general, becomes illiquid. Similarly, we expect basis trade arbitrageurs to prefer, all else being equal, basis trades with bonds whose trading costs covary less with the overall bond market illiquidity. Thus, we expect a negative coefficient in the cross-sectional regression of bases on bond liquidity betas.

The bond liquidity market beta (denoted as $\beta_{Liq,M}$) is the comovement between bond i's illiquidity and the market return, where the market return is measured by the CRSP value-weighted stock market return. As in Acharya and Pedersen (2005), we expect arbitrageurs to prefer at the margin negative basis trades for bonds that tend to have trading costs that covary more with market returns. Indeed, bonds with high trading costs in market downturns potentially lead to lower profits upon an unwind precisely in bad states. Thus, we expect, all else being equal, a less negative basis on bonds with larger bond liquidity market betas (i.e., a positive regression coefficient).

TABLE 3 Limit-to-arbitrage risk factors

Panel A. Summary statistics						
Variable	**Notation**	**Mean**	**Std**	**Med**	**P10**	**P90**
Lending Fee	*Fee*	0.12	0.40	0.09	0.04	0.16
Credit Rating	*Rat*	12.64	3.53	13.50	7.00	16.50
Bond Illiquidity (level)	*Liq*	0.42	1.68	0.11	0.00	0.88
Bond Liquidity Risk	β_{Liq}	0.62	2.72	0.21	−0.67	2.28
Bond Liquidity Mkt Risk	β_{LiqM}	−0.03	0.36	−0.01	−0.19	0.13
Counterparty Risk	β_{CP}	0.13	0.29	0.05	0.00	0.36
Funding Liquidity Risk	β_{FL}	−0.03	1.32	−0.03	−0.68	0.69

Panel B. Correlation in the noncrisis period							
	Fee	***Rat***	***Liq***	β_{Liq}	β_{LiqM}	β_{CP}	β_{FL}
Fee	1.00						
Rat	−0.29	1.00					
Liq	−0.01	−0.05	1.00				
β_{Liq}	0.00	−0.04	0.02	1.00			
β_{LiqM}	0.00	0.02	−0.07	−0.08	1.00		
β_{CP}	0.21	−0.31	0.03	0.00	0.00	1.00	
β_{FL}	−0.01	0.00	0.00	0.00	0.00	0.03	1.00

Panel C. Correlation in the crisis period							
	Fee	***Rat***	***Liq***	β_{Liq}	β_{LiqM}	β_{CP}	β_{FL}
Fee	1.00						
Rat	−0.20	1.00					
Liq	−0.04	−0.02	1.00				
β_{Liq}	0.00	−0.05	0.27	1.00			
β_{LiqM}	0.03	0.07	−0.38	−0.39	1.00		
β_{CP}	0.03	−0.03	0.03	0.02	0.02	1.00	
β_{FL}	0.00	0.05	−0.03	−0.01	0.08	−0.06	1.00

Notes: This table provides the summary statistics of limit-to-arbitrage factors in Panel A and their correlation values in Panel B, which is further divided into the crisis period in Panel C (July 2007–September 2008) and the noncrisis period (July 2006–June 2007 and October 2009–December 2014).

The three measures provide a characterization of bond liquidity risk. Intuition suggests that the three measures should be correlated. In Table 3, we demonstrate that the correlation between liquidity (risk) measures is low during the noncrisis period, ranging from 2% to 8% in the absolute value. However, the correlations increase to 27–39% during the crisis period.

4.3 | Counterparty risk

Counterparty risk became a primary concern facing participants in the financial markets during the 2007–2009 crisis. Counterparty risk is the risk that the protection seller, typically a broker dealer, cannot make good on its commitment to the protection buyer in the case of default. Thus, counterparty risk should make the insurance less valuable and lower the CDS spread, possibly contributing to the negative basis. As previously explained, the higher the correlation between the default events of the underlying entity and the protection seller, the larger the expected counterparty risk.[12] The challenge is how to measure the correlation between the default risk of the underlying entity and the counterparty selling the CDS protection.[13]

The CDS market is over-the-counter and the exact nature of counterparties is not known. Furthermore, the process of netting makes it difficult to establish an aggregate measure of counterparty risk for individual reference entities.[14]

To establish a measure of the counterparty risk faced by an investor engaging in a specific basis trade, we note that throughout our sample, the primary dealers trade, on average, more than 90% of the total transaction dollar volume. Thus, it seems reasonable to construct a counterparty risk measure for a representative CDS issuer using the list of primary dealers designated by the Federal Reserve Bank of New York.[15] These primary dealers are banks and security broker-dealers that trade in US government securities with the Federal Reserve System. To become qualified as a primary dealer, a firm must be in compliance with capital standards under the Basel Capital Accord with at least $100 million of Tier I capital for a bank or above $50 million of regulatory capital for a broker-dealer. As trading partners of the central bank, these primary dealers often are the biggest and most competitive financial institutions who happen to be the dominant issuers of CDS contracts. As of September 2008, there were 19 primary dealers including Citigroup, J.P. Morgan Chase, and Goldman Sachs (the complete list can be found in Appendix B). The list changes over time as some primary dealers may fail to meet the required capital standards. Accordingly, we update the components of the primary dealer index. For example, the index includes Lehman Brothers' Holdings prior to its bankruptcy on September 15, 2008, but excludes it afterward and adds Nomura Securities International, Inc., starting from July 27, 2009.

For the primary dealer index, we calculate its CDS spread weighted by each constituent's market capitalization. As illustrated in Figure 2, the primary dealers have a striking increase in their default risk during the financial crisis of 2008–2009 and the peak of the European sovereign debt crisis of 2011–2012. A direct implication is that the insurance contracts sold by these dealers should be less valuable and the protection buyers face higher counterparty risk. We measure an underlying entity's counterparty risk as the beta coefficient of regressing the change of firm-level CDS to the change of the primary dealer CDS:

$$\beta_{i,cp} = \frac{cov\left(\Delta CDS_i, \Delta CDS_{index}\right)}{var\left(\Delta CDS_{index}\right)}. \tag{6}$$

The higher the β_{cp}, the greater the likelihood of a joint default and the less valuable we expect the protection to be when purchased from that counterparty. Thus, we expect a negative coefficient in the cross-sectional regression of the bases on counterparty betas.

4.4 | Funding liquidity risk

For an arbitrageur entering a basis trade, the primary risk is that the basis becomes more negative at the same time as their funding costs widen. Thus, we proxy funding liquidity risk by the regression coefficient of the change in the basis on a measure of the change in funding costs or funding liquidity premium.

The literature has considered many proxies to measure the funding liquidity premium including the Libor–OIS spread, the TED spread (Libor–TBill), the Repo–TBill spread, and the OIS–TBill spread. One concern with the Libor-indexed spreads is that they are contaminated by financial intermediary credit risk and could be correlated with counterparty risk, especially during a financial crisis. For this reason, we follow Nagel (2016) and use the three-month Repo–TBill spread. Bai, Krishnamurthy, and Weymuller (2018) propose an alternative measure of the OIS–TBill spread. Both the Repo–TBill spread and the OIS–TBill spread have similar time-series patterns, and they have a correlation value of 0.90. All of our empirical results remain unchanged in magnitude and significance if we use the OIS–TBill spread as the alternative proxy of funding the liquidity premium.

Our estimate of the funding liquidity beta is therefore defined as:

$$\beta_{i,fl} = \frac{\text{cov}\left(\Delta Basis_i, \Delta Repo - TBill\right)}{\text{var}\left(\Delta Repo - TBill\right)}. \tag{7}$$

The lower the funding liquidity beta, the less aggressively an arbitrageur would invest in that basis trade as the basis will become more negative when their funding costs increase. We expect a positive coefficient in the cross-sectional regression of the bases on funding liquidity betas.

4.5 | Summary

We summarize the expected signs of the cross-sectional determinants of the CDS-bond basis in the following table based on our previous discussion.

Factors	Expected Cross-sectional Correlation with Negative Basis
Lending fee	+
Credit rating	+
Bond illiquidity	–
Bond liquidity risk	–
Bond liquidity mkt risk	+
Counterparty Risk	–
Funding liquidity risk	+

Panel B of Table 3 presents the correlation values of the limit-to-arbitrage factors. In the noncrisis period, the correlations among collateral quality proxies, bond trading liquidity, counterparty risk, and funding liquidity risk are trivial. For example, funding liquidity beta is almost uncorrelated with any other factors with correlation values between −0.01 and 0.03. Bond illiquidity or liquidity risk also has a negligible correlation with the other factors. In the crisis period, the correlation between bond illiquidity and bond liquidity risk increases, but the correlation values among other risk factors remain small. The low correlations relieve the concern of colinearity across the limit-to-arbitrage risk factors.

5 | CROSS-SECTIONAL DETERMINANTS OF THE CDS-BOND BASIS

We now investigate the cross-sectional variation in the CDS-bond basis using Fama and MacBeth (1973) regressions. To mitigate the impact of noise in the daily data, we adopt the weekly frequency throughout our empirical analysis, where the bond-level weekly basis is the average value of the daily basis within each week for a particular bond. For the limit-to-arbitrage factors except collateral quality, all of the beta coefficients are estimated for each bond at each week using a rolling window of the past 60-week observations.[16] For collateral quality, we calculate the weekly averages of the daily corporate bond loan lending fee and of its credit rating.

5.1 | Bond-level results

We report the Fama–MacBeth (1973) regression results for four subperiods: Before Crisis, Crisis I, Crisis II, and Postcrisis. In detail, weekly cross-sectional regressions are run for the following specification and nested versions thereof:

$$\begin{aligned} Basis_{i,t} &= \alpha_t + \gamma_{t,Rat} Rat_{i,t} + \gamma_{t,Fee} Fee_{i,t} + \gamma_{t,Liq} Liq_{i,t} + \gamma_{t,Bliq} \beta_{i,t,Liq} + \gamma_{t,BliqM} \beta_{i,t,LiqM} \\ &+ \gamma_{t,CP} \beta_{i,t,CP} + \gamma_{t,FL} \beta_{i,t,FL} + \varepsilon_{i,t}, \ \forall t, \end{aligned} \tag{8}$$

where $Basis_{i,t}$ is the CDS-bond basis on bond i in week t. Rat is the bond credit rating in numeric values, Fee is the bond lending fee, Liq is the bond trading illiquidity level, β_{Liq} denotes bond liquidity risk, β_{LiqM} denotes bond liquidity market risk, and β_{CP} and β_{FL} denote the counterparty risk and funding liquidity risk. All of themeasures are defined in Section 4. We standardize the explanatory variables cross-sectionally each week. As such, the coefficients are comparable across variables and over sample periods.

Table 4 reports the time-series average of the estimated coefficients γ_ts and the average adjusted R^2 values over the 466 weeks from July 2006 to December 2014 for the negative CDS-bond basis. The t-statistics based on Newy–West adjusted standard errors are given in squared brackets.

Focusing on the multivariate regressions in the last column, as the results remain largely similar to the univariate case, we find that collateral quality, bond illiquidity, counterparty risk, and funding liquidity risk all enter significantly, especially during the crisis period, achieving an R^2 of 36.6%. The sign of the significant coefficients is consistent with our expectations summarized in Section 4.5. During the peak of the crisis from September 2008 to September 2009, the most important factors driving the cross-sectional differences in basis are credit ratings with an estimated coefficient of 1.902 (t-stat = 6.78), bond illiquidity with an estimated coefficient of −0.676 (t-stat = −3.16), funding liquidity risk with an estimated coefficient of 0.794 (t-stat = 3.71), and counterparty risk with an estimated coefficient of −0.545 (t-stat = −5.22). In the first stage of the crisis from July 2007 to August 2008, the same set of factors has significant explanatory power, though to a relatively smaller degree. For example, the coefficient for credit rating drops from 1.902 in Crisis II to 0.570 in Crisis I, the coefficient for counterparty risk drops from −0.545 to −0.181, and funding liquidity risk drops from 0.794 to 0.132.

However, during the noncrisis period (both before and after the crisis), only credit rating remains consistently significant in explaining the cross-sectional variations of the basis, indicating that collateral quality is always relevant for a basis arbitrage trade. The estimated coefficients for counterparty risk are also statistically significant in the noncrisis period, but have much smaller magnitude (the coefficient is 0.059 before and 0.163 after the crisis). In addition, the signs of the counterparty risk coefficient in the noncrisis period are positive. This is counter to our expectation and is likely due to the strong colinearity of counterparty risk with credit ratings and lending fees (the correlations are −0.31 and 0.21, respectively). Finally, the bond liquidity factors (both illiquidity and liquidity risk) lose explanatory power in the noncrisis period.

There is one surprising finding. Lending fees have a significant, but negative coefficient in the cross-sectional regressions throughout the crisis and noncrisis periods. This is at variance with our expectations. Our interpretation is that lending fees also proxy for the short interest in a bond. That is, a high lending fee not only arises when a bond is highly valued as collateral and is in high demand in the securities lending market (in this case, the bond has a low repo rate), but may also arise when a bond is being heavily shorted and is in scarce supply in the securities lending market. If the lending fee during the crisis primarily captures short interest, then the negative sign on lending fees could reflect the fact that arbitrageurs refrain from entering a basis trade on the related bond that is heavily shorted. In effect, arbitrageurs may worry about "catching a falling knife" when entering such a basis trade as the negative market view on the bond signaled by the high short interest suggests that the basis could diverge even further in the near future.

In sum, the empirical model is reasonably successful in explaining the cross-sectional variation in the bases. All of the factors (except for lending fee) have the signs as expected, consistent with the hypothesis that the marginal investor acting as a leveraged hedge fund trade-off risk and return when allocating scarce risk capital to different basis investment opportunities.

5.2 | Portfolio-level results

To minimize the noise in the bond-level cross-sectional regression results, we also consider running the Fama and MacBeth (1973) regression at the portfolio level. We first examine the distribution of bond size throughout the sample and categorize bonds into four bins by the 25th, 50th, and 75th percentile. Within each bin of bond size, we further categorize bonds into another six bins by their credit ratings: AAA/AA, A, BBB, BB, B, CCC&Below. In so doing, we construct the time series of 24 corporate bond portfolios sorted by size and rating. In Panel A of Table 5, we report the average basis of the 24 portfolios. The basis is clearly monotonically increasing with bond credit quality and also monotonically decreasing with size as one might have expected.

In Panel B, we present the portfolio-level cross-sectional regression results. We find a very high R^2 of around 80% for the bases of the 24 portfolios. Most coefficients are statistically significant throughout the subsample periods. The signs on credit rating, bond illiquidity and liquidity risk, counterparty risk, and funding liquidity risk are also consistent

TABLE 4 Multivariate Fama–MacBeth regression of the negative CDS-bond basis

	Panel A. Before Crisis (July 2006–June 2007)					Panel B. Crisis I (July 2007–Aug. 2008)				
Lending Fee	0.000				0.000	−0.005***				−0.003***
	[−1.499]				[0.718]	[−4.613]				[−3.008]
Credit Rating	0.070***				0.074***	0.622***				0.570***
	[7.245]				[3.863]	[6.793]				[7.246]
Bond Illiquidity (level)		0.000**			−0.020		−0.004			−0.236
		[−2.055]			[−1.453]		[−1.225]			[−0.850]
Bond Liquidity Risk		0.003			0.001		−0.345***			−0.465***
		[0.114]			[0.080]		[−2.861]			[−4.096]
Bond Liquidity Mkt Risk		−0.004			−0.022		−0.822			−0.807*
		[−0.237]			[−1.166]		[−1.586]			[−1.921]
Counterparty Risk			0.000***		0.059***			−0.003***		−0.181***
			[3.051]		[4.363]			[−4.508]		[−2.563]
Funding Liquidity Risk				0.000	[0.019]				0.001	0.132**
				[1.158]	[1.038]				[1.598]	[2.336]
Adj. R^2	0.147	0.101	0.056	0.022	[0.356]	0.263	0.113	0.057	0.043	0.408

(Continues)

TABLE 4 (Continued)

	Panel C. Crisis II (Sept. 2008–Sept. 2009)					Panel D. Postcrisis (Oct. 2009–Dec. 2014)				
Lending Fee	−0.004***				−0.004***	−0.002***				−0.001***
	[−2.689]				[−2.578]	[−5.640]				[−5.007]
Credit Rating	2.124***				1.902***	0.739***				0.779***
	[7.771]				[6.780]	[27.213]				[22.580]
Bond Illiquidity (level)		−0.009***			[−0.676]		−0.001***			0.002
		[−4.123]			[−3.156]		[−2.816]			[0.107]
Bond Liquidity Risk		0.538**			0.101		−0.001			0.006
		[2.344]			[0.391]		[−0.065]			[0.508]
Bond Liquidity Mkt Risk		1.269***			0.517**		−0.017			0.033*
		[4.725]			[1.722]		[−1.494]			[2.172]
Counterparty Risk			−0.007***		−0.545***			−0.002***		0.163***
			[−6.033]		[−5.216]			[−7.714]		[8.313]
Funding Liquidity Risk				0.010***	0.794***				0.000	0.039
				[5.302]	[3.706]				[0.752]	[0.767]
Adj. R^2	0.215	0.075	0.037	0.080	0.366	0.287	0.022	0.025	0.145	0.442

Notes: This table reports the average coefficients from the Fama–MacBeth (1973) cross-sectional regressions of the CDS-bond bases on risk factors including collateral quality proxies, such as lending fees and credit ratings, three proxies of bond trading liquidity and liquidity risk, counterparty risk, and funding liquidity risk. The sample is limited to negative basis only. The regression is conducted at the weekly frequency for 444 weeks from July 2006 to December 2014. Except collateral quality, all of the other factors are calculated for each bond at each week using a rolling window of the past 60-week observations. The weekly lending fee or credit rating is calculated as the weekly average of the daily observations. Credit rating is a series of numeric values with one referring to CCC, 2 to CC-,… and 21 to AAA. The *t*-statistics based on Newey–West (1987) adjusted standard errors are given in squared brackets. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

TABLE 5 The CDS-bond basis portfolios sorted by size and rating

Panel A. Average basis (bps) for portfolios sorted by size and rating						
	AAA/AA	A	BBB	BB	B	CCC&Below
Small	−126	−125	−201	−258	−318	−534
2	−79	−103	−146	−229	−283	−487
3	−75	−105	−132	−240	−285	−436
Large	−79	−98	−130	−238	−266	−631

Panel B. Fama–MacBeth regression results				
	Before Crisis	Crisis I	Crisis II	Post-Crisis
Lending Fee	0.000	−0.007***	−0.005***	−0.004***
	[1.371]	[−4.927]	[−3.224]	[−6.431]
Credit Rating	0.111***	0.542***	1.268***	0.702***
	[−5.431]	[12.294]	[10.039]	[35.092]
Bond Illiquidity (level)	−0.083***	0.013	−0.261	0.261***
	[−2.650]	[0.081]	[−1.119]	[5.831]
Bond Liquidity Risk	−0.053	−0.622***	−0.421	0.312***
	[−1.007]	[−5.803]	[−1.430]	[4.691]
Bond Liquidity Mkt Risk	0.020	−0.503***	0.809***	0.000
	[0.517]	[−2.491]	[2.506]	[0.040]
Counterparty Risk	0.145***	−0.281***	−0.800***	0.118***
	[5.598]	[−3.924]	[−6.425]	[2.450]
Funding Liquidity Risk	−0.011	0.111	0.720***	0.156***
	[−0.371]	[1.150]	[3.431]	[2.696]
Adj. R^2	0.799	0.759	0.774	0.839

Notes: Each week, we construct corporate bond portfolios based on bonds outstanding amounts and bond ratings. Bond size is divided into four categories based on the 25th, 50th, and 75th percentiles of the whole sample, and the bond rating is categorized into six bins including AAA/AA, A, BBB, BB, B, and CCC&Below. There are a total of 24 portfolios. The sample is limited to negative basis only. Panel A reports the average basis value (bps) for each portfolio. Panel B presents the average coefficients from the Fama–MacBeth (1973) cross-sectional regressions of the CDS-bond bases for 24 portfolios on risk factors including collateral quality, three proxies of bond trading liquidity and liquidity risk, counterparty risk, and funding liquidity risk. The regression is conducted at the weekly frequency for 444 weeks from July 2006 to December 2014. Except collateral quality proxies, all of the other factors are calculated for each bond at each week using a rolling window of the past 60-week observations. The weekly collateral quality is calculated as the weekly average of the daily corporate bond loan lending fee. The *t*-statistics based on Newey–West (1987) adjusted standard errors are given in squared brackets. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

with our expectations during the crisis. However, we note that outside of the crisis period, counterparty risk and bond illiquidity remain significant, but, surprisingly, their signs flip likely due to the correlation with other risk factors.

6 | CONCLUSION

We have analyzed the cross-sectional variation in the CDS-bond basis during the financial crisis. Focusing on the cross section of the CDS-bond basis is interesting as it provides a natural testing ground for the literature that models limits to arbitrage and the behavior of arbitrageurs with limited capital facing multiple arbitrage opportunities.

We find that most notably during the post-Lehman crisis period, several limit-to-arbitrage measures, such as collateral quality, bond illiquidity and liquidity risk, counterparty risk, and funding liquidity risk can explain a significant fraction of the cross-sectional variation in the CDS-bond basis. After the crisis, the explanatory power of risk

measures decreases substantially, but collateral quality proxies, such as credit ratings and lending fees, remain significant. Interestingly, throughout the sample, bond lending fees remain highly negatively related to the bond basis. We interpret this as evidence that arbitrageurs refrain from entering basis trades that have large trading frictions (i.e., that are costly to fund or to trade) associated with them and especially when there is a large amount of short interest in the bond (signaled by a high lending fee), indicating that the basis could further diverge in the short run.

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NOTES

[1] For example, investment grade corporate credit spreads, as measured by the CDX.IG Index, rose from 50 basis points (bps) in early 2007 to more than 250 bps at the end of 2008. Even at the safest end of the spectrum, the widening was dramatic. AAA-rated synthetic debt products, which would have been deemed virtually risk free before the crisis, saw their spreads widen dramatically. The CDX.IG super senior tranche widened from 5 to 100 bps, CMBX AAA "super-duper" widened from 2 to 700 bps, and ABS-HEL AAA tranche prices rose from 0% to 20% upfront plus 500 bps running. These numbers illustrate that it became much more expensive to insure AAA-rated debt across various markets (corporate, residential, and commercial real estate).

[2] In the 2003 definition, the International Swap and Derivative Association (ISDA) lists six items as credit events: (a) bankruptcy, (b) failure to pay, (c) repudiation/moratorium, (d) obligation acceleration, (e) obligation default, and (f) restructuring. For more detail, see "2003 ISDA Credit Derivatives Definitions," released on February 11, 2003. Also see Duffie and Singleton (2003) for a detailed description.

[3] Duffie (1999) discusses the specific conditions and demonstrates why this relation might not exactly hold in practice.

[4] We assume, conservatively, that the investor does not earn any interest on the posted margin. If the investor was paid interest on the margin, then the margin would be funded at $Libor + f_m$, where f_m would be the funding spread net of the interest earned on the posted margin.

[5] We use the approximation $\Delta B^{mid} \approx -D_B(\Delta Libor + \Delta YS^{mid})$, where YS is the bond yield credit spreads. We size the position in the CDS so that $N_{CDS}D_{CDS} = D_B$. Further, we define the CDS bid–ask spread $BAS_t^C = 2(CDS_t^{ask} - CDS^{mid}) = (CDS^{ask} - CDS^{bid})$. Then, we assume $B_t^{ask} - B_t^{bid} \approx D_B * BAS_t^B$. Thus, $B_{t+1}^{bid} - B_t^{ask} = -D_B(\Delta Libor + \Delta YS^{mid} + 0.5\Delta BAS_t^B)$. Putting this all together, we derive the above expression with $BAS_t = 0.5(BAS_t^B + BAS_t^C)$ as the average of the bid–ask spread on the bond and on the CDS.

[6] There is not an unanimously agreed day for the beginning of the subprime crisis. Popular opinion is that the subprime crisis started in August 2007. Here, we take a conservative stance by starting the crisis period in July 2007. Also, it is unclear to the earliest ending date of the U.S. financial crisis. According to the NBER business cycle dates, the recession ended by June 2009. We allow for another three months in the Crisis II period.

[7] Bonds also contain other option features, such as putable, redeemable/callable, exchangeable, and fungible. With the exception of callable bonds, bonds with other option features are a relatively small portion of the sample. However, callable bonds constitute about 67% of the entire sample. Thus, we keep the callable bonds in our final sample, but we also conduct robustness checks for a smaller sample filtering out those bonds with option features.

[8] Under the 2003 Credit Definitions by the International Swap and Derivative Association (ISDA), there are four types of restructuring clauses: cumulative restructuring (CR), MR, modified–modified restructuring (MM), and no restructuring (XR). MR is used by most broker-dealers in the US market. This convention holds until April 8, 2009. Afterward, the US market adopts the XR convention. For consistency, we choose the MR documentation clause throughout our sample.

[9] Overtime, there are also a lot of variations in the basis in a way that cannot solely be explained by the variation in the collateral funding spread and, as we argue below, is unlikely to be explained solely by changes in haircuts.

[10] The general model in Garleanu and Pedersen (2011) predicts that other factors (such as the covariance of the underlying cash flows with aggregate consumption) in addition to the margin differential should predict the difference in the basis. It is only for the specific application to the CDS basis that they focus on the margin difference. Our data suggest that it is important to look for additional factors.

[11] There is no agreement on the best measure for corporate bond liquidity. Thus, we also construct alternative liquidity measures, such as the bid–ask price spread and the Amihud (2002) measure. These measures generate similar empirical results.

[12] That counterparty risk is not irrelevant can be seen from the Lehman Brothers case. Suppose an investor had purchased protection on Washington Mutual from Lehman Brothers. Washington Mutual defaulted only a few days after Lehman. Without marking to market, the investor would be a regular claimant in bankruptcy for the protection purchased from Lehman leading to, at best, a partial loss. Of course, if ISDA agreements were well enforced and provided that the investor had negotiated full-two-way mark-to-market with Lehman, the risk would be further mitigated. However, in practice, it is likely that most funds would have ended with at least some partial loss as a result of this double default.

[13] Arora, Gandhi, and Longstaff (2012) look for counterparty fixed effects using a proprietary data set of CDS transaction prices by 14 CDS dealers selling credit protection on one underlying firm to identify the counterparty risk component of CDS spreads.

[14] In September 2008, the bankruptcy of Lehman Brothers caused almost $400 billion to become payable to the buyers of CDS protection referenced against the insolvent bank. However, the net amount that changed hands was around $7.2 billion. This difference is due to the process of "netting." Market participants cooperated so that CDS sellers were allowed to deduct from their payouts the funds due to them from their hedging positions. Dealers generally attempt to remain risk neutral so that their losses and gains after big events will, on the whole, offset each other.

[15] Data source: http://www.newyorkfed.org/markets/pridealers_current.html.

[16] A bond is included in our sample if it has at least 24 weekly basis observations in the 60-week rolling window prior to the test week. Our data start from July 2001 and we report regression results since July 2006. Our results are also robust to different rolling windows in estimating the limit-to-arbitrage risk factors. That is: (a) a 36-week rolling window instead of 60 weeks using weekly data or (b) a 60-day rolling window using daily data.

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APPENDIX A: THE PAR EQUIVALENT CDS METHODOLOGY

We present the PECDS methodology developed by J.P. Morgan to calculate the CDS-bond basis in Section 2. This survival-based valuation approach provides an apples-to-apples measure across the cash-bond spread and the CDS spread.

The fair value of the coupon on a CDS is set so that the expected present value of the premium leg is equal to the expected present value of the contingent payment (Duffie, 1999). Assuming that we have a zero coupon discount curve $Z(t)$ extracted from swap spreads and assuming a constant intensity survival probability $S(t)$, the expected present value of the premium leg is given by:

$$PV_{\text{premium}}(C) = \sum_{i=1}^{n} Z(t_i)S(t_i)C^* dt + \sum_{i=1}^{n} Z\left(\frac{t_i + t_{i+1}}{2}\right)[S(t_{i-1}) - S(t_i)]C^* dt/2, \tag{A.1}$$

where the second component is the present value of the accrued interest upon default (assumed to occur halfway between t_{i-1} and t_i). The expected present value of the contingent leg is:

$$PV_{\text{contingent}} = (1 - R)\sum_{i-1}^{n} Z\left(\frac{t_i + t_{i+1}}{2}\right)[S(t_{i-1}) - S(t_i)], \tag{A.2}$$

where R is the recovery rate. The fair CDS spread is the number C that sets:

$$PV_{\text{premium}}(C) = PV_{\text{contingent}}. \tag{A.3}$$

The PECDS uses the market price of a bond to calculate a spread based on CDS-implied default probabilities. First, we need to derive a CDS-implied default probability curve $S_{CDS}(t_i)$ by sequentially plugging in CDS spread with maturity

from 1 year to 10 years. Then, we must obtain a bond-implied survival probability curve $S_{bond}(t_i)$. Using the CDS-implied survival probability as a prior, we calculate the bond-implied survival probability curve as the one that minimizes the pricing error between the market price and derived bond price:

$$S_{bond}\left(t_i\right) = S_{CDS}\left(t_i\right) + \varepsilon, \tag{A.4}$$

$$s.t.\ \varepsilon = arg\ min\left(PV\left(S_{bond}\right) - Market\ Price\ of\ Bond\right)^2. \tag{A.5}$$

Then, the bond-implied CDS spread term structure is defined by substituting the survival probability term structure fitted from the bond prices, $S_{bond}(t)$, into the following equation for PECDS spreads, denoted as PECDS:

$$PECDS = \frac{(1-R)\sum_{i=1}^{n} Z\left(\frac{t_i+t_{i+1}}{2}\right)[S_{bond}(t_{i-1}) - S_{bond}(t_i)]}{\sum_{i=1}^{n}\left[z(t_i)S_{bond}(t_i) * dt + Z\left(\frac{t_i+t_{i+1}}{2}\right)[S_{bond}(t_{i-1}) - S_{bond}(t_i)] * \frac{dt}{2}\right]}. \tag{A.6}$$

APPENDIX B: THE PRIMARY DEALERS LIST

Below is the list of primary dealers as of July 27, 2009 reported on the website of the Federal Reserve Bank of New York.

BNP Paribas Securities Corp.
Banc of America Securities LLC
Barclays Capital Inc.
Cantor Fitzgerald & Co.
Citigroup Global Markets Inc.
Credit Suisse Securities (USA) LLC
Daiwa Securities America Inc.
Deutsche Bank Securities Inc.
Goldman, Sachs & Co.
HSBC Securities (USA) Inc.
Jefferies & Company, Inc.
J. P. Morgan Securities Inc.
Mizuho Securities USA Inc.
Morgan Stanley & Co. Incorporated
Nomura Securities International, Inc.
RBC Capital Markets Corporation
RBS Securities Inc.
UBS Securities LLC.

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